REVOFin

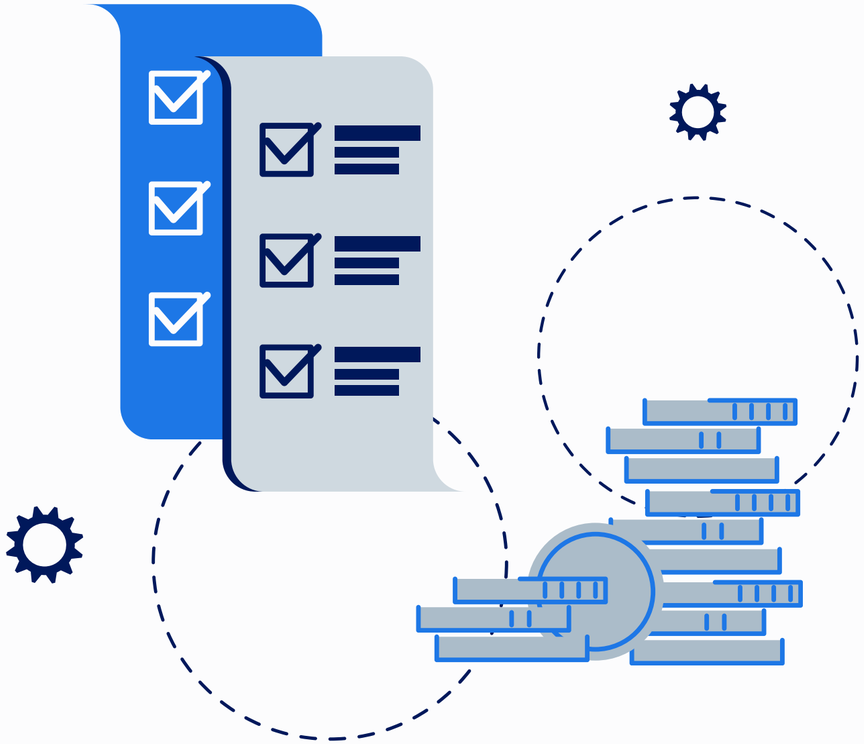
Data-Driven Lending Optimization & Customer Insights

Problem Statement

RevoFin lacks clear visibility into portfolio risk distribution, with potential exposure to high delinquency (DPD > 30 days) segments that may negatively impact TKB30 and overall portfolio performance.

Objective

Quantify portfolio health by measuring Outstanding (OS), ENR, and TKB30, identify high-risk segments with elevated DPD > 30 rates, and provide actionable recommendations to improve TKB30 and reduce default exposure.



```sql
WITH cleaned_customer AS (
  SELECT
    *,
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r'[a-zA-Z0-9]'), '') AS clean_customer_id
  FROM `nextproject-001.FIP.customer`
),

cleaned_loan AS (
  SELECT
    *,
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r'[a-zA-Z0-9]'), '') AS clean_customer_id
  FROM `nextproject-001.FIP.loan`
),

active_loans AS (
  SELECT *
  FROM cleaned_loan
  WHERE loan_status IN ('Current', 'In Grace Period', 'Late', 'Default')
),

loan_with_geo AS (
  SELECT
    l.*,
    r.region,
    r.subregion
  FROM active_loans l
  LEFT JOIN `nextproject-001.FIP.state_regions` r
    ON l.state = r.state
),

```

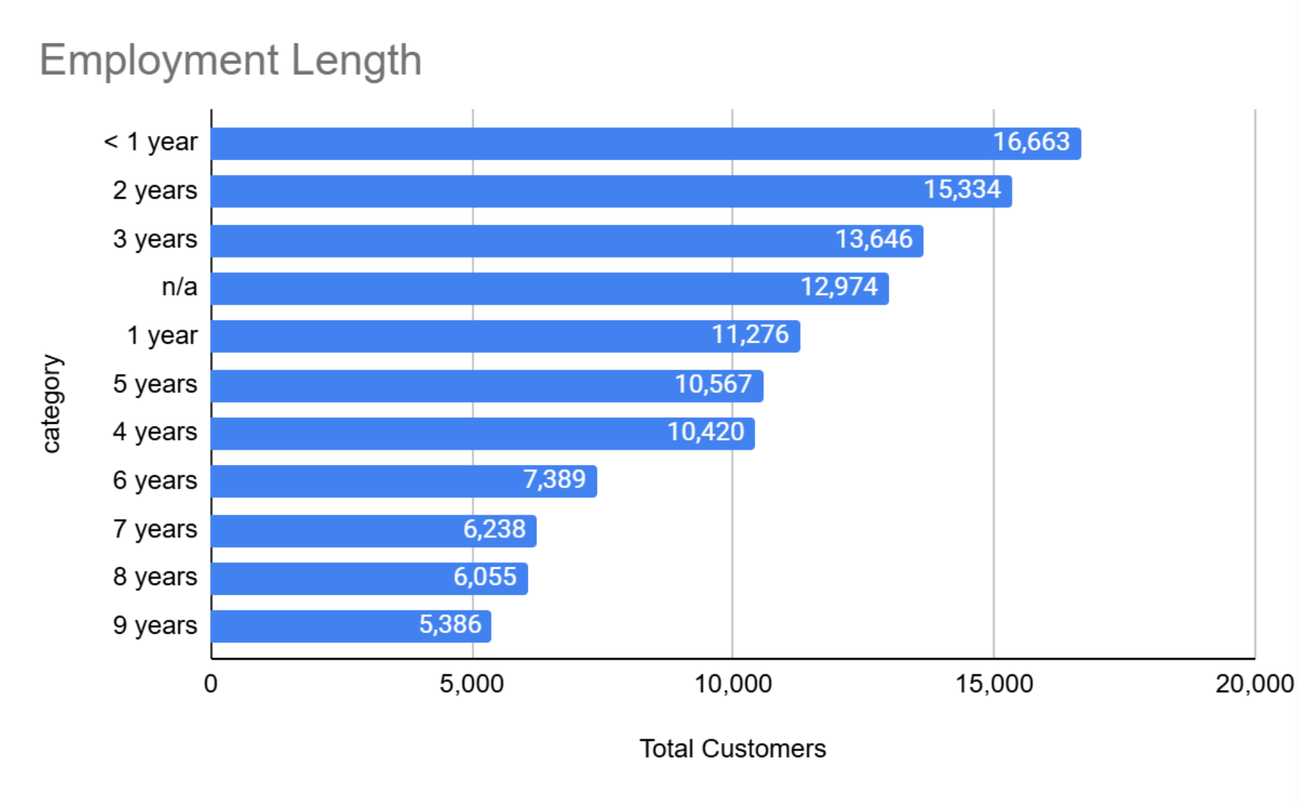
```sql
final_table AS (
  SELECT
    l.*,
    c.emp_length,
    c.home_ownership,
    c.annual_inc,
    c.verification_status
  FROM loan_with_geo l
  LEFT JOIN cleaned_customer c
    ON l.clean_customer_id = c.clean_customer_id
)

SELECT
  COUNT(*) AS total_rows,
  COUNT(DISTINCT loan_id) AS total_loans,
  COUNT(DISTINCT clean_customer_id) AS total_customers,
  MIN(issue_date) AS earliest_loan_date,
  MAX(issue_date) AS latest_loan_date
FROM final_table
```

Row	total_rows	total_loans	total_customers	earliest_loan_date	latest_loan_date
1	172111	172111	172111	April 2013	September 2019

Cleaned Dataset Overview

The dataset contains 172K loan records with a one-to-one mapping between loans and customers, covering the period from April 2013 to September 2019, providing a consistent and reliable foundation for portfolio analysis over time.

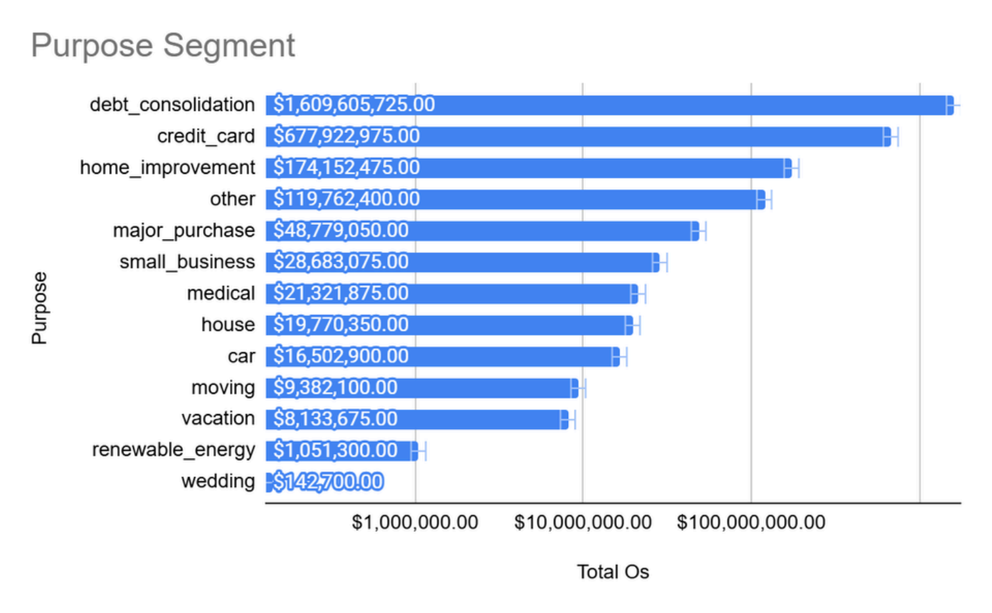
Customer Risk Profile

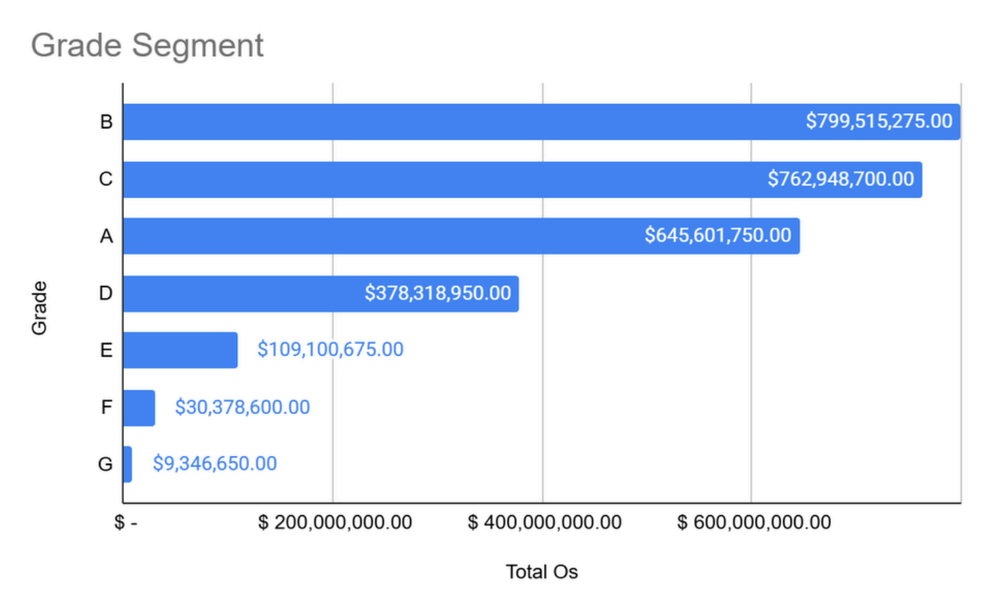


Category	Segment	Total Customers
Income Group	High Income	170873
	Medium Income	1139
	Low Income	99

- The portfolio is highly concentrated in low-risk borrowers (~56K with >10 years of employment, ~170K classified as high income)
→ indicates strong **selection bias**
- Most customers have ongoing financial obligations (~84K with mortgages, ~67K renting) → implies continued cash flow pressure
- Around ~16K borrowers have employment tenure <1 year
→ potential early risk signals (income stability not yet proven)
- Approximately ~67K customers remain unverified
→ suggests gaps in underwriting and verification processes
- Overall performance appears strong, but there are hidden risks from:
  - concentration in “safe” borrower profiles
  - unverified customers
  - short-tenure segments

Loan Portfolio Structure





$1.6B (Debt Consolidation)

- Portfolio is concentrated in mid-risk borrowers (Grades A–C), indicating a balanced risk-return profile
- Interest rates are aligned with risk grades, showing effective risk-based pricing
- Over 75% of loans are used for debt refinancing, highlighting dependency on financially leveraged borrowers
- Longer-term loans (60 months) offer higher yields but introduce greater risk exposure
- Overall portfolio reflects a trade-off between growth and stability

TKB 30 - Portofolio Health

```sql
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    *,
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r'[a-zA-Z0-9]'), '') AS clean_customer_id
  FROM `nextproject-001.FIP.loan`
),

active_loans AS (
  SELECT *
  FROM cleaned_loan
  WHERE loan_status IN ('Current', 'In Grace Period', 'Late', 'Default')
),

final_table AS (
  SELECT *
  FROM active_loans
)

SELECT
  SUM(funded_amount) AS portfolio_os,

  -- ENR (exclude default)
  SUM(CASE WHEN loan_status != 'Default' THEN funded_amount ELSE 0 END)
  / SUM(funded_amount) AS ENR,

  -- TKB30 (default sebagai proxy DPD>30)
  1 - (
    SUM(CASE WHEN loan_status = 'Default' THEN funded_amount ELSE 0 END)
    / SUM(funded_amount)
  ) AS TKB30

FROM final_table
```

portfolio_os	ENR	TKB30
$ 2.7 B	99 %	99 %

The portfolio shows near-perfect performance, with ENR and TKB30 both close to 100%, indicating minimal observed default exposure. However, this result is likely driven by strong selection bias and limited risk variation within the dataset, rather than reflecting true portfolio risk. Therefore, the current metrics should be interpreted with caution.

“The portfolio appears healthy, but not necessarily resilient.”

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Insights

- Near-perfect performance (ENR & TKB30 ~100%) driven by limited risk exposure
- Strong selection bias toward high-income, low-risk borrowers
- High dependency on refinancing loans (>75% debt-related purposes)
- Limited risk visibility due to lack of delinquency variation

Recommendations

- Expand risk monitoring by incorporating early delinquency indicators (e.g., late payments) to better capture emerging risk
- Reassess portfolio diversification by gradually testing expansion into broader customer segments while maintaining risk controls
- Strengthen underwriting controls, particularly around verification processes, to mitigate hidden risk exposure
- Enhance data collection (e.g., DPD, repayment behavior) to enable more accurate and forward-looking risk assessment
- Continuously monitor cohort performance to detect early signs of deterioration as the portfolio matures